

Adarsh Patel

Software Engineer | Full Stack | Backend | Python | AI Systems

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TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C++

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend & APIs: Node.js, Express.js, Flask, REST APIs, MQTT, Microservices

Databases: PostgreSQL, MongoDB, InfluxDB, Prisma ORM

DevOps & Cloud: Docker, GitHub Actions, AWS, Azure Digital Twins, Linux, Grafana

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, Distributed Systems

EXPERIENCE

Planarc — Full Stack Engineer Intern

Jul 2025 – Nov 2025

- Delivered **25+ reusable React components** across **4 core product flows** following OOP and component-driven design patterns, cutting duplicated UI code by **~40%**.
- Optimized **lazy-loading and bundle-splitting**, reducing initial page load from **~3.2s to 2.4s (25% improvement)** on mid-range devices.
- Refactored **3 backend API modules** and authored API contracts in an Agile environment, reducing cross-team integration bugs noticeably.

PROJECTS

VibeCoder — Cloud-Based Collaborative AI IDE

[GitHub](#)

Next.js 15 · React 19 · TypeScript · WebContainers API · Monaco Editor · Gemini API · Prisma · MongoDB

- Engineered a **browser-based cloud IDE** using Next.js 15 and WebContainers, supporting **6 development environments** (React, Next.js, Express, Vue, Hono, Angular), achieving **~15.4s cold boot** for container setup and **5ms warm-session reconnect latency**.
- Architected a **context-aware AI code suggestion system** using Gemini 1.5 Flash with custom prompt engineering, reducing autocomplete latency from **5–15s with local LLM fallback to under 0.8s**, a **90%+ speed improvement**.
- Built a **real-time terminal error scanner and AI Wrench** to detect and auto-patch runtime/compile errors; implemented secure public playground sharing, an **AI README Generator parsing up to 20 workspace files**, and a persistent **6-theme Monaco editor system**.

NoiseSense — Distributed Acoustic Intelligence & Noise Mapping System

[GitHub](#)

Flutter · Python · Flask · MQTT · YAMNet · InfluxDB · Docker · Gemini AI · Grafana

- Built a **real-time acoustic intelligence system** achieving **85%+ classification accuracy** across tested environments, integrating YAMNet for **521-class sound recognition** with **5 DSP acoustic features** extracted from GPS-tagged audio.
- Engineered an **asynchronous MQTT-based processing pipeline** achieving **3–6 sec ML processing per sample** and **<50 ms MQTT latency** under tested strong Wi-Fi conditions, integrating FFmpeg preprocessing, ML inference, and InfluxDB time-series storage.
- Developed **GPS-based noise maps and heatmaps** with offline-first mobile collection achieving **100% recovery of 8/8 queued samples and zero data loss during offline testing**; integrated Grafana monitoring and **Gemini-powered automated intelligence reports**.

Smart Industry 4.0 — Digital Twin Based Predictive Maintenance

[GitHub](#)

Python · WebSockets · SQLite · Scikit-learn · Three.js · React · Azure Digital Twins · Arduino · ESP8266

- Architected a **real-time Industrial IoT digital twin** integrating Arduino/ESP8266 sensor nodes with **Python asyncio WebSockets**, **5-second heartbeat polling**, auto-reconnect logic, SQLite persistence, and **Azure Digital Twins** for live factory monitoring.
- Integrated an online **IsolationForest anomaly detection model** using travel time and vibration signals, self-training after a **minimum 50-sample window** with a **5% contamination threshold** to enable zero-label predictive maintenance and real-time fault detection.
- Built an interactive **Three.js 3D factory digital twin** using a **~7.9 MB GLB model** synchronized with live WebSocket sensor states; developed React/Chart.js monitoring, historical telemetry and Excel/JSON export. Ranked **Top 24 out of 8,000+ projects** at UIP-2025.

ACHIEVEMENTS

- Odoo Hackathon 2025** — Finalist, **Top 78 out of 19,000+ participants**; built [TurfSpot](#), a sports venue booking platform ([Live](#) | [GitHub](#)).
- Unisys Innovation Program 2025 (UIP-2025)** — Ranked **Top 24 out of 8,000+ projects**.
- LeetCode** — Solved **400+ problems** across arrays, trees, graphs, dynamic programming, binary search, and sliding window.

EDUCATION

Manakula Vinayagar Institute of Technology, Puducherry

Oct 2022 – May 2026

B.Tech — Electronics and Communication Engineering

CGPA: 7.7 / 10

Vivekanandha Higher Secondary School

2020 – 2022 | Class XII: 85% | Class X: 81%